TRANSFORMING CANCER DIAGNOSIS: MULTI-CANCER DETECTION USING VISION TRANSFORMERS AND AI

A PROJECT PRELIMINARY REPORT

Submitted by

ALAN THOMAS(MBC22CS0015)
MATHEWS VINOY(MBC22CS047)
PRINCE J(MBC22CS054)
SANDESH V ANTONY(MBC22CS058)
SIRIL T NINAN(MBC22CS061)

to

APJ Abdul Kalam Technological University

In partial fulfillment of requirements for the award of the degree

of

Bachelor of Technology

in

Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE

OCTOBER 2025

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MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY

2025-2026

CERTIFICATE

This is to certify that this report entitled **"TRANSFORMING CANCER DIAGNOSIS: MULTI-CANCER DETECTION USING VISION TRANSFORMERS AND AI"** submitted by **ALAN THOMAS** (MBC22CS015), **MATHEWS VINOY** (MBC22CS047), **PRINCE J** (MBC22CS020), **SANDESH V ANTONY** (MBC22CS036), **SIRIL T NINAN** (MBC22CS061), and to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering, is a bonafide report of the project work carried out by them under the guidance and supervision of **Dr. Ushus Maria Joseph**, Associate Professor, Department of Computer Science and Engineering. This report in any form has not been submitted to any other University or Institute for any purpose.

Dr. Ushus Maria Joseph	**Dr. Ushus Maria Joseph**
Project Guide	Head of the Department
Associate Professor	Associate Professor
Department of CSE	Department of CSE

DECLARATION

We hereby declare that this report entitled **"TRANSFORMING CANCER DIAGNOSIS: MULTI-CANCER DETECTION USING VISION TRANSFORMERS AND AI"** is a bonafide record of the preliminary project work carried out by us under the supervision of **Dr. Ushus Maria Joseph** in the Department of Computer Science and Engineering Mar Baselios Christian College of Engineering and Technology, Peermade.The project proposal presented in this report have been presented for the award of any other degrees.

Peermade

Date: 25 October 2025

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COLLEGE VISION & MISSION

Vision

An Engineering Institute with global quality to groom competent engineers equipped to address the changing needs of society.

Mission

Our efforts are dedicated for developing a learner centric education environment to:

- Provide value-based technical learning
- Practice real world problem solving
- Foster team work in engineering design
- Inspire innovations and R&D

DEPARTMENT VISION & MISSION

Vision

To become an excellent department by empowering the graduates to address the challenges in the fast-changing scenario, adhering to ethical values.

Mission

- Inculcate skills to solve complex engineering problems and instill entrepreneurial capabilities.
- Equip the students to advance with the recent trends and technology through Research and Innovations in the field of computer science.
- Facilitate the development of professional behaviour, ethical values and the spirit of service to the society.

PROGRAMME EDUCATIONAL OBJECTIVES

Our Graduates will be able to:

PEO1: Think creatively and critically to solve problems in a global context with varying complexity.

PEO2: Exhibit leadership qualities, Ethical Values, managerial and entrepreneurship skills.

PEO3: Contribute inventions and innovative advancements in information technology through life-long learning in a sustainable manner.

POs & PSOs

PROGRAM OUTCOMES (POs)

Engineering Graduates will be able to:

- **PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering Fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2. Problem Analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- **PO3. Design/ Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- **PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- **PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- **PO6. The Engineer and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- **PO7. Environment and Sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- **PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- **PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- **PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

- **PO11. Project Management and Finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

- **PO12. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

PROGRAM SPECIFIC OUTCOMES (PSOs)

- **PSO1:** Apply theoretical and practical knowledge to analyse and model hardware systems with varying complexity.

- **PSO2:** Utilize modern programming platforms, tools and logical reasoning skills to analyse and develop application software.

- **PSO3:** Apply computational and networking skills to solve and develop secure communication environment.

Course Objectives

- To apply engineering knowledge in practical problem solving.

- To foster innovation in design of products, processes or systems.

- To develop creative thinking in finding viable solutions to engineering problems.

Course Outcomes (COs)

After successful completion of the course, the students will be able to:

CO1	Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: **Apply**).
CO2	Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: **Apply**).
CO3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: **Apply**).
CO4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: **Apply**).

CO5	Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: **Analyze**).
CO6	Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: **Apply**).

Mapping of Course Outcomes with Program Outcomes

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO1	2	2	2	1	2	2	2	1	1	1	1	2
CO2	2	2	2		1	3	3	1	1		1	1
CO3							3	2	2	1		
CO4					2		3	2	2	3	2	
CO5	2	3	3	1	2							1
CO6					2			2	2	3	1	1

ABSTRACT

Cancer remains one of the leading causes of mortality worldwide, and early detection is critical for effective treatment. Traditional diagnosis methods, reliant on manual interpretation of medical images, can be time-consuming, inconsistent, and prone to error. This project explores an automated, scalable solution for detecting multiple types of cancer using Vision Transformers (ViT)—a cutting-edge architecture in deep learning that has shown superior performance in image analysis tasks. By leveraging the self-attention mechanism of ViTs, the system captures both local features and global dependencies within high-resolution medical images such as histopathological slides, CT scans, and dermoscopy images. The proposed framework is designed as a multi-class classification model capable of detecting various cancer types, including breast, lung, skin, and brain cancer. Extensive preprocessing and data augmentation techniques enhance training efficiency, while optimization methods such as transfer learning and fine-tuning address the challenges of limited medical datasets. The solution, deployed via a web-based interface built with Flask and integrated with PyTorch models, demonstrates improved accuracy, generalizability, and clinical relevance compared to traditional CNN-based approaches. This project underscores the potential of Vision Transformers in revolutionizing cancer diagnosis through automated, reliable, and rapid assessments across diverse imaging modalities..

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Chapter 1
INTRODUCTION

Cancer remains one of the most critical global health challenges, contributing to a significant percentage of deaths each year. Early diagnosis is the cornerstone of effective treatment and improved survival rates. However, conventional cancer detection techniques—such as manual image interpretation, microscopic examination, and laboratory report analysis—are often time-consuming, subjective, and prone to human error. The emergence of Artificial Intelligence (AI) has opened new possibilities in medical diagnostics, offering tools that can process vast amounts of medical data with speed, precision, and consistency.
This project, titled "Transforming Cancer Diagnosis: Multi-Cancer Detection Using Vision Transformers and AI," aims to develop an integrated AI-driven diagnostic system capable of detecting and interpreting cancer from multiple medical data sources. The system leverages state-of-the-art deep learning architectures in both Computer Vision and Natural Language Processing (NLP) to enhance diagnostic accuracy and clinical decision-making. The proposed framework is structured into three main components, each employing a specialized AI model:
—

1.1 Vision Transformer (ViT) – Medical Image Classification Model

The first component utilizes the Vision Transformer (ViT) architecture for cancer image classification. Unlike traditional Convolutional Neural Networks (CNNs), ViT divides an image into fixed-size patches and processes them using transformer encoders. This mechanism enables the model to capture long-range dependencies and contextual relationships across the entire image. The ViT model is trained on datasets containing medical images such as MRI, CT, and histopathological slides to detect abnormal growth patterns indicative of cancer. By learning discriminative features and spatial patterns, the ViT model can classify images into various categories such as cancerous and non-cancerous with high precision. This phase forms the visual foundation of the system, enabling automated, consistent, and scalable image-based cancer detection.

1.2 Fine-Tuned Large Language Model (LLM) – Clinical Text Understanding

The second component employs a fine-tuned Large Language Model (LLM), such as Med-PaLM, GPT, or Gemma, to process and interpret medical text data. This model is fine-tuned using domain-specific datasets that include clinical notes, pathology reports, radiology summaries, and cancer-related research literature. The goal of this phase is to empower the LLM with medical reasoning capabilities, allowing it to perform specialized tasks such as summarizing patient reports, identifying relevant biomarkers, interpreting diagnostic results, and assisting in treatment recommendations. Through fine-tuning, the model learns to understand complex medical terminology and correlate textual descriptions with diagnostic outcomes, bridging the gap between human clinical language and AI comprehension.

1.3 Blood Report Analysis Model – Predictive Cancer Detection

The third component focuses on developing a Blood Report Analysis Model to determine the likelihood of cancer based on a patient's blood test results. The model processes numerical and textual data extracted from laboratory reports, such as hemoglobin levels, white blood cell counts, and other relevant biomarkers. Using machine learning techniques, the system analyzes these parameters to detect irregular patterns that may indicate the presence of cancer. This phase provides an additional layer of diagnostic validation by integrating biochemical data, enhancing the overall reliability of the cancer detection framework.

By combining the strengths of Vision Transformers for visual analysis, fine-tuned Large Language Models for text comprehension, and blood report–based predictive modeling for biological assessment, this system offers a comprehensive and intelligent approach to multi-cancer detection. The integration of these components into a unified framework ensures faster diagnosis, improved accuracy, and scalable implementation across healthcare settings. Ultimately, this project aims to bridge the gap between advanced AI research and real-world clinical application, providing medical professionals with a powerful tool for early cancer detection and patient care.

Chapter 2
LITERATURE SURVEY

2.1 AI-Enhanced Lung Cancer Prediction: A Hybrid Model's Precision Triumph (2025)

Authors: Cyrille Yetuyetu Kesiku and Begonya Garcia-Zapirain

The paper "AI-Enhanced Lung Cancer Prediction: A Hybrid Model's Precision Triumph" by Cyrille Yetuyetu Kesiku and Begonya Garcia-Zapirain proposes a hybrid deep learning model combining Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) with an attention mechanism for early lung cancer detection using patients' medical notes.
The model was trained on the MIMIC-IV dataset and achieved 98.1% accuracy and 96.2% MCC, outperforming LSTM and BioBERT models. By integrating CNN's local feature extraction and BiLSTM's contextual understanding, the model effectively captures complex relationships in medical text, showing strong potential for precision medicine and automated clinical diagnosis.

Advantages

- **Hybrid Model Excellence:**Combines CNN for spatial feature extraction and BiLSTM with attention for contextual understanding, leading to superior classification accuracy.
- **High Performance:** Achieved an accuracy of 98.1% and an MCC of 96.2%, outperforming leading architectures like BioBERT and LSTM..
- **Generalizability:** Demonstrated consistent results not only in the medical domain but also on non-medical text datasets, confirming robustness and adaptability.

Disadvantages

- **Dependence on Text Data:** The system focuses primarily on textual medical notes rather than integrating multimodal data such as images or lab results, which may limit holistic diagnosis.
- **High Computational Demand:** The hybrid CNN-BiLSTM-Attention model, despite being efficient, still requires substantial computational resources for training and deployment.
- **Dataset-Specific Bias:** Reliance on MIMIC-IV data (mostly from a single hospital network) may affect model generalizability to other clinical settings or patient demographics.

2.2 Early Detection of Pancreatic Cancers Using Liquid Biopsies and Hierarchical Decision Structure (2022)

Authors: Deepesh Agarwal, Obdulia Covarrubias-Zambrano, Stefan H. Bossmann, and Balasubramaniam Natarajan

Summary
This research introduces a novel machine learning-based framework for the early detection of pancreatic cancer using liquid biopsies. The authors combine ultrasensitive nanobiosensors that detect protease and arginase enzyme activity with an information fusion–based hierarchical decision structure (HDS) to identify pancreatic cancer at its localized stage.
The study models the detection task as a multi-class classification problem and proposes both Hard (HDS) and Soft (SDS) decision structures. While HDS performs stepwise classification for "healthy," "localized," and "metastatic" cases, SDS enhances interpretability by providing confidence probabilities for each decision. The proposed model achieved an accuracy of 92%, significantly outperforming conventional multi-class classification models (74%). Despite using a relatively small dataset (31 patients and 48 healthy controls), the method demonstrates strong potential for early and non-invasive cancer detection through simple blood tests.

Advantages

- **Non-Invasive Liquid Biopsy**: The proposed model uses liquid biopsies instead of invasive tissue sampling, allowing early pancreatic cancer detection through simple blood tests, which is safer and more patient-friendly.
- **High Diagnostic Accuracy**: The Hierarchical Decision Structure (HDS) achieved a remarkable 92% accuracy, outperforming conventional multi-class classification models and ensuring more reliable diagnostic results.
- **Confidence-Based Prediction**: The Soft Decision Structure (SDS) provides class probabilities that indicate confidence levels for each prediction, helping clinicians assess the reliability of results before diagnosis.
- **Targeted Feature Engineering**: By weighting and selecting only significant biomarkers such as protease and arginase, the model improves learning efficiency and reduces overfitting on small datasets.

Disadvantages

- **Limited Dataset Size**:The research was based on only 79 subjects (31 cancer and 48 healthy), making it difficult to generalize results to larger or more diverse patient populations.
- **Cancer-Specific Focus:**: The model is specifically designed for pancreatic cancer and cannot be directly extended to other cancer types without retraining and modification.
- **High Dependence on Sample Quality**:Performance relies heavily on the accuracy and consistency of blood samples, meaning errors in collection or handling can affect prediction results.

- **Model Complexity**:The hierarchical framework, involving multiple classifiers and feature-engineering steps, increases computational complexity and training time.

2.3 An Explainable Artificial Intelligence Model for the Classification of Breast Cancer (2025)

Authors: Tarek Khater, Abir Hussain, Riyad Bendardaf, Iman M. Talaat, Hissam Tawfik, Sam Ansari, and Soliman Mahmoud

Summary
This paper presents an explainable artificial intelligence (XAI) model for classifying breast cancer as benign or malignant using two well-known datasets — Wisconsin Breast Cancer (WBC) and Wisconsin Diagnostic Breast Cancer (WDBC). The authors trained several machine learning algorithms (KNN, ANN, SVM, RF, XGBoost) and achieved the highest accuracy of 97.7% (KNN) and 98.6% (ANN). The study emphasizes model interpretability through Permutation Importance, Partial Dependence Plot (PDP), and SHAP methods, identifying the "Bare Nuclei" and "Worst Area" features as key indicators of malignancy. This research highlights the importance of transparency in AI-driven healthcare systems to support clinical trust and early cancer detection.

Advantages

- **High Accuracy Classification**:The model achieves 97.7–98.6% accuracy, showing excellent reliability for early breast cancer detection using clinical data.
- **Explainable Predictions**:Through SHAP, PDP, and Permutation Importance, the system provides clear visual explanations for each decision, improving transparency for clinicians.
- **Feature Importance Discovery**:The study identifies Bare Nuclei and Worst Area as the two most influential biomarkers, enhancing medical understanding of tumor behavior.
- **Dual Dataset Integration**:By linking both WBC and WDBC datasets, the model validates cross-dataset consistency and strengthens diagnostic robustness.

Disadvantages

- **Limited Dataset Diversity**:The study relies solely on the Wisconsin datasets, which are well-known but not highly diverse, limiting real-world generalization.
- **High Computational Cost**: XAI methods like SHAP and PDP are computationally intensive, increasing analysis time and limiting scalability for large datasets.
- **Restricted Feature Visualization**:PDP visualizations can only handle two features simultaneously, limiting insight into complex multi-feature interactions.

2.4 Improved Water Strider Algorithm with Convolutional Autoencoder for Lung and Colon Cancer Detection on Histopathological Images (2024)

Authors: Hamed Alqahtani, Eatedal Alabdulkreem, Faiz Abdullah Alotaibi, Mrim M. Alnfiai, Chinu Singla, and Ahmed S. Salama

Summary
This paper proposes an Improved Water Strider Algorithm with Convolutional Autoencoder (IWSACAE-LCCD) for detecting lung and colon cancers from histopathological images. The system integrates median filtering for noise removal, MobileNetV2 for feature extraction, Improved Water Strider Algorithm (IWSA) for hyperparameter optimization, and a Convolutional Autoencoder (CAE) for final classification. Tested on the LC25000 dataset, the model achieved an accuracy of 99.41%, outperforming methods like ResNet50, Faster R-CNN, and BERTL-HIALCC. The approach ensures faster execution (1.57s) and high precision in classifying five cancer subtypes, highlighting its capability for real-time and accurate cancer diagnostics.

Advantages

- **High Classification Accuracy**: The proposed IWSACAE-LCCD model achieved a 99.41% accuracy, outperforming other deep learning models in lung and colon cancer detection.
- **Optimized Feature Extraction**: By combining MobileNetV2 with the Improved Water Strider Algorithm, the model automatically fine-tunes hyperparameters for better feature extraction and robustness.
- **Noise Reduction Efficiency**: The median filtering technique effectively removes salt-and-pepper noise, improving the clarity and quality of histopathological images before classification.
- **Lightweight and Fast Performance**:MobileNetV2 ensures low computational cost and high processing speed, achieving an execution time of just 1.57 seconds per sample.

Disadvantages

- **High Computational Resource Requiremen**: Training the CAE and MobileNetV2 models with IWSA optimization demands powerful GPUs and extended processing time.
- **Dataset Limitation**: The model is tested only on the LC25000 dataset, which, while comprehensive, may not fully represent real clinical variations in histopathology images.
- **Limited Cancer Scope**:The proposed framework is designed exclusively for lung and colon cancers, requiring modifications to handle other cancer types.

2.5 A Novel Deep Learning Approach for Accurate Cancer Type and Subtype Identification (2024)

Authors: Jabed Omar Bappi, Mohammad Abu Tareq Rony, Mohammad Shariful Islam, Samah Alshathri, and Walid El-Shafai

Summary
This study introduces a novel deep learning framework capable of classifying eight main cancer types and twenty-six subtypes using over 130,000 images from Kaggle datasets. The model integrates hybrid CNN–LSTM architectures, introducing two new models—Vception (VGG + Inception) and Vmobilenet (VGG + MobileNet)—combined with an X-OR gate fusion layer for improved prediction confidence. Two training strategies were used: one performing main and sub-class classification together, and another hierarchical method with PCA-KNN refinement for lymphoma. The model achieved exceptional performance, reaching 99.95% accuracy for main classes and 99.13% for sub-classes, outperforming state-of-the-art methods.

Advantages

- **Multi-Cancer Coverage**: Unlike most prior works that target single cancer types, this model simultaneously detects eight cancer types and 26 subtypes, enhancing diagnostic versatility.
- **Hybrid Model Innovation**: Introduces Vception and Vmobilenet, merging strengths of VGG, Inception, and MobileNet with LSTM layers to improve feature learning and temporal understanding
- **X-OR Fusion Accuracy Boost:**: The novel X-OR gate integration significantly reduces misclassifications by combining outputs from two models, improving final prediction reliability.
- **High Precision and Recall**: The framework achieves 99.25% main-class accuracy and 97.8% subclass accuracy, setting a new benchmark for deep learning-based cancer detection.

Disadvantages

- **High Computational Complexity**: The multi-stage CNN-LSTM-KNN-XOR fusion requires large GPU resources and longer training times compared to simpler architectures.
- **Dataset Restriction**: The study uses Kaggle datasets only, which may not represent real clinical diversity or unseen cancer morphologies.
- **Overfitting Risk**: Due to dataset similarity, deep hybrid models can overfit training data, reducing adaptability to real-world medical images.
- **Complex Workflow Design**: The multi-step pipeline (preprocessing, hybrid CNN, XOR logic, PCA-KNN) complicates implementation in hospital environments.
- **Limited Explainability**: Although accurate, the architecture lacks explicit explainable AI (XAI) visualization tools for clinician interpretation.

2.6 MedViT: A Robust Vision Transformer for Generalized Medical Image Classification (2023)

Authors:Omid Nejati Manzari, Hamid Ahmadabadi, Hossein Kashiani, Shahriar B. Shokouhi, and Ahmad Ayatollahi

Summary
This paper introduces MedViT, a hybrid CNN–Transformer architecture that integrates the local feature extraction power of CNNs with the global attention capability of Vision Transformers (ViTs) for medical image classification. MedViT employs an Efficient Convolution Block (ECB) and a Local Transformer Block (LTB) to capture both short-term and long-term dependencies, enhancing its ability to analyze complex medical images across multiple modalities such as X-ray, CT, ultrasound, and OCT. It also introduces a unique Patch Momentum Changer (PMC) augmentation technique to improve robustness against adversarial attacks. Tested on MedMNIST-2D datasets (12 medical datasets), MedViT achieved up to 94.2% AUC and 85.1% accuracy, outperforming CNNs like ResNet and hybrid baselines while maintaining lower computational cost.

Advantages

- **Hybrid Architecture Efficiency**: MedViT effectively combines CNN locality and Transformer global modeling, capturing detailed and contextual features for superior medical image understanding.
- **High Classification Performance:**:Achieved up to 94.2% AUC and 85.1% accuracy across 12 datasets, surpassing traditional CNN and AutoML methods in medical diagnostics.
- **Adversarial Robustness**: The Patch Momentum Changer (PMC) improves model resilience against attacks (FGSM, PGD), ensuring reliability in real clinical use.

Disadvantages

- **Dataset Dependence**: Performance is validated only on MedMNIST datasets, limiting verification on real clinical or high-resolution datasets.
- **Limited Real-World Testing**: The model remains experimentally validated, lacking trials on real hospital data or diverse demographic samples.
- **Potential Overfitting Risk**: Despite regularization, hybrid ViTs may overfit smaller datasets, reducing effectiveness in unseen data distributions.
- **Interpretability Challenges**: While Grad-CAM provides some insights, full explainability of transformer decisions remains difficult for medical experts.
- **Training Cost & Time**: The hierarchical structure and self-attention layers increase training duration and GPU usage, hindering fast experimentation.
- **Data Quality Sensitivity**: Accuracy depends heavily on clean, standardized images; poor quality or noisy scans can degrade prediction accuracy.

2.7 Improved EATFormer: A Vision Transformer for Medical Image Classification (2024)

Authors:Yulong Shisu, Susano Mingwin, Yongshuai Wanwag, Zengqiang Chenso, and Sunshin Huing

Summary
This paper presents an Enhanced Evolutionary Algorithm-based Transformer (EATFormer) architecture for medical image classification. The model combines Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to extract both local and global visual patterns, improving accuracy and efficiency. It introduces three major innovations: the Multi-Scale Region Aggregation (MSRA) module for capturing multi-receptive-field features, the Global and Local Interaction (GLI) module for balancing fine and broad information, and the Modulated Deformable Multi-Head Self-Attention (MD-MSA) module for handling irregular spatial locations. Tested on Chest X-ray and Kvasir datasets, the model achieved 95.33% and 94.37% accuracy, surpassing well-known CNNs and Transformers such as DenseNet, Swin, and MedViT, while maintaining efficient computation.

Advantages

- **Hybrid CNN–Transformer Integration**: Combines CNN's local perception with Transformer's global attention, providing a balanced and powerful image representation framework.
- **Multi-Scale Feature Learning**: The MSRA module captures patterns from multiple receptive fields, enhancing model robustness across different image resolutions.
- **Enhanced Spatial Awareness**: The MD-MSA module dynamically models irregular spatial structures, improving feature extraction in complex medical images.
- **Superior Classification Accuracy**: Achieves 95.3% accuracy on Chest X-ray and 94.4% on Kvasir, outperforming Swin Transformer and MedViT baselines.

Disadvantages

- **High Computational Requirements**: The hybrid structure with multiple modules (MSRA, GLI, MD-MSA) requires strong GPU resources and longer training time.
- **Dataset Limitation**: Evaluations are performed only on Chest X-ray and Kvasir datasets, which restricts generalization across other modalities.
- **Architectural Complexity**: The presence of several new modules increases implementation difficulty and model tuning overhead.
- **Limited Real-World Validation**: No clinical or real-time testing is conducted; results are restricted to controlled experimental datasets.

2.8 Comparative Analysis of Multi-Omics Integration Using Graph Neural Networks for Cancer Classification (2017)

Authors:Fadi Alharbi, Aleksandar Vakanski, Boyu Zhang, Murtada K. Elbashir, and Mohanad Mohammed

Summary
This study introduces a multi-omics integration approach using Graph Neural Networks (GNNs) for accurate cancer classification. It compares Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), and Graph Transformer Networks (GTN) applied to mRNA, miRNA, and DNA methylation data. The proposed models—LASSO-MOGCN, LASSO-MOGAT, and LASSO-MOGTN—use LASSO regression for feature selection and employ correlation matrices and Protein–Protein Interaction (PPI) networks to capture biological relationships. The LASSO-MOGAT model achieved the highest accuracy of 95.9This demonstrates that integrating multi-omics data using GNN architectures enhances cancer subtype identification and biomarker discovery.

Advantages

- **Multi-Omics Integration Power**:Combining mRNA, miRNA, and DNA methylation data enables comprehensive biological insight, improving cancer subtype classification.
- **Graph-Based Learning Efficiency**:Utilizing GNNs (GCN, GAT, GTN) effectively models complex biological networks, outperforming traditional deep learning techniques.
- **Superior Accuracy**: The LASSO-MOGAT model achieved 95.9% accuracy, setting a new benchmark in multi-omics cancer classification.
- **Correlation-Based Graph Strength**: Correlation matrices enhance graph construction, enabling better detection of shared cancer-specific gene signatures.

Disadvantages

- **High Computational Demand**: Integrating large-scale omics data and training multiple GNN architectures require significant GPU resources.
- **Complex Implementation**: The pipeline—including LASSO selection, graph construction, and training—is technically challenging and time-consuming.
- **Limited Dataset Diversity**: Evaluation was restricted to Pan-Cancer Atlas data, lacking external validation on independent datasets.

2.9 Classification of Cancer Types Using Graph Convolutional Neural Networks (2020)

Authors:Ricardo Ramirez, Yu-Chiao Chiu, Allen Hererra, Milad Mostavi, Joshua Ramirez, Yidong Chen, Yufei Huang, and Yu-Fang Jin

Summary
This paper introduces a Graph Convolutional Neural Network (GCNN) framework for classifying 33 cancer types and normal tissues using gene expression data from the TCGA dataset. Four GCNN variants were built using different graph structures: co-expression, co-expression + singleton, PPI, and PPI + singleton networks. The PPI + singleton GCNN achieved the highest accuracy of 94.7The study also employed in silico gene perturbation analysis to identify 428 cancer-specific marker genes, confirming that the GCNN effectively distinguishes cancer-specific patterns rather than tissue-specific ones.

Advantages

- **Graph-Based Learning Strength**: GCNN models leverage biological gene–gene relationships instead of treating features as independent, improving prediction accuracy.
- **High Multi-Cancer Accuracy**: Achieved 94.7% accuracy across 33 cancer types and normal samples, surpassing previous CNN and deep learning models.
- **Cancer-Specific Marker Identification**: Discovered 428 discriminative genes that contribute to cancer classification, enhancing biological interpretability.
- **Integration of Multiple Graph Types**: Combining PPI and co-expression graphs provides complementary biological insights, improving overall performance.

Disadvantages

- **High Computational Complexity**: Training GCNNs on large-scale omics data demands significant GPU resources and long processing times.
- **Limited to TCGA Dataset:**: The study relies only on TCGA data, lacking external or real-clinical validation for generalization.
- **Model Interpretability Challenge**:Despite perturbation analysis, understanding latent graph features remains difficult for medical interpretation.

2.10 Deep Residual Learning for Image Recognition (2015)

Authors:Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun

Summary
This groundbreaking paper introduces ResNet (Residual Networks), a deep learning architecture designed to overcome the degradation problem encountered in very deep neural networks. The key idea is to use residual connections (skip connections), allowing layers to learn residual functions rather than direct mappings, making it easier to optimize

very deep models. ResNet achieved 3.57% top-5 error on the ImageNet 2015 classification task and won 1st place in the ILSVRC 2015 competition. With up to 152 layers, ResNet demonstrated that deeper architectures can be efficiently trained while maintaining lower computational complexity than VGG networks. The framework has since become a foundation for modern computer vision, enabling advancements in object detection, segmentation, and transfer learning.

Advantages

- **Overcomes Degradation Problem**: ResNet solves the issue where deeper models perform worse by introducing skip connections, making gradient flow more stable.
- **Enables Ultra-Deep Architectures**: The design allows networks of over 1000 layers to train effectively without performance saturation or exploding gradients.
- **High Accuracy**: Achieved 3.57% error on ImageNet, significantly surpassing previous CNN architectures like VGG and GoogLeNet.
- **Residual Mapping Efficiency**: Learning residuals instead of full mappings simplifies optimization, leading to faster convergence and better generalization.
- **Computationally Efficient**: Despite its depth, ResNet-152 requires fewer computations than VGG-19, thanks to optimized bottleneck blocks.

Disadvantages

- **High Computational Demand**:Training deep ResNets requires powerful GPUs, long training times, and large memory resources.
- **Complex Hyperparameter Tuning**: Depth selection, learning rate scheduling, and initialization parameters require careful tuning to avoid suboptimal convergence.
- **Overfitting on Small Datasets**:Extremely deep architectures may overfit when trained on limited or imbalanced data without regularization.
- **Interpretability Issues**: Although effective, the internal behavior of residual layers remains hard to interpret for human understanding.

2.11 Densely Connected Convolutional Networks (DenseNet) (2017)

Authors:Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger

Summary
This paper proposes DenseNet, a deep learning architecture where each layer is directly connected to all subsequent layers to enhance information flow and gradient propagation. Unlike traditional CNNs or ResNets that rely on additive skip connections, DenseNet uses feature-map concatenation, promoting feature reuse and reducing redundancy. The model significantly mitigates the vanishing gradient problem, improves parameter efficiency, and achieves state-of-the-art performance on datasets like CIFAR-10, CIFAR-100, SVHN, and ImageNet with fewer parameters than ResNet. DenseNet's architecture leads

to implicit deep supervision, compact feature representation, and high generalization performance across visual recognition benchmarks.

Advantages

- **Enhanced Gradient Flow**: Dense connectivity ensures better gradient propagation, solving vanishing gradient issues common in very deep networks.
- **Feature Reuse Efficiency**: By concatenating feature maps, each layer can reuse features from all preceding layers, reducing redundancy and improving learning efficiency.
- **High Parameter Efficiency**: DenseNets achieve top-tier accuracy with significantly fewer parameters than ResNet and VGG architectures.
- **Improved Generalization**: Feature reuse and implicit deep supervision enhance model regularization, leading to strong generalization on unseen data.

Disadvantages

- **High Memory Usage**: Dense connectivity increases GPU memory requirements due to concatenation of feature maps across layers.
- **Implementation Complexity**:The dense inter-layer connections make the architecture complex to design and fine-tune.
- **Training Overhead**: Although parameter-efficient, DenseNet's training time can be longer due to numerous layer interconnections.
- **Scalability Constraints**: For extremely deep architectures, concatenation may lead to memory bottlenecks and slower data transfer.

2.12 Multi-Cancer Detection Using CNN, InceptionV3, and Vision Transformer (ViT) (2025)

Authors: Saleha Habeeba and Dr. Khaja Mahabubullah

Summary

This paper presents an AI-assisted diagnostic framework that combines CNN, InceptionV3, and Vision Transformer (ViT) architectures for multi-cancer classification. The system classifies seven cancer types — brain, breast, oral, lung, cervical, lymphoma, and kidney — using medical and histopathological images. Trained using a supervised learning pipeline with preprocessing, augmentation, and stratified sampling, the models were evaluated using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. The ViT model achieved the highest performance with an average accuracy of 94.44The model was deployed using Streamlit, enabling real-time image upload and diagnosis, and achieved under 1.2 seconds of inference time per image.

Advantages

- **Hybrid Deep Learning Approach**: Combines the feature extraction strength of CNNs, the multi-scale learning of InceptionV3, and the global attention of ViT for superior accuracy.

- **High Multi-Cancer Accuracy**: The ViT-based system achieved 94.44% average accuracy, outperforming CNN and InceptionV3 across all cancer types.
- **Real-Time Deployment**:Implemented with Streamlit, allowing clinicians to upload medical images and receive instant diagnostic feedback.
- **Robust Data Preprocessing**:Strict inclusion and exclusion criteria ensured high-quality training data, improving model reliability and reducing bias.
- **Strong Evaluation Framework**: Used five-fold cross-validation with minimal standard deviation ($\pm$1.5%), confirming model stability and generalization.

Disadvantages

- **Limited Dataset Size**:The study relies on publicly available datasets, which may not represent real clinical diversity or rare cancer subtypes.
- **Lack of Explainability**: Despite high accuracy, the model offers limited interpretability, making it hard for clinicians to understand prediction rationale.
- **High Computational Demand**: Training ViT models requires substantial GPU power and memory, limiting accessibility in low-resource settings.
- **Dataset Imbalance Risk**: Class imbalance among cancer types can bias the model toward dominant classes.

2.13 CONVMCD: Convolutional Neural Network for Multi-Cancer Early Detection (2023)

Authors:Ashifur Rahman, Md Shakhawat Hossain, M. M. Mahbubul Syeed, Kaniz Fatema, and Mohammad Faisal Uddin

Summary
The paper proposes an automated multi-cancer early detection (MCED) system based on Convolutional Neural Networks (CNNs) for identifying lung and colon cancers from digital histopathological images. The authors compared four CNN architectures—MobileNetV3, VGG16, DenseNet201, and ResNet50—to determine the most efficient model for classification. The system was trained using the Kaggle Multi-Cancer Dataset, which includes five cancer subtypes (colon_aca, colon_bnt, lung_aca, lung_bnt, lung_scc). Experimental results showed that MobileNetV3 achieved the best performance with 99.66% accuracy, 99% precision, and 99% sensitivity, outperforming all other tested architectures. The research highlights the potential of AI-assisted diagnostic systems in reducing manual effort, bias, and diagnostic time in cancer screening.

Advantages

- **High Accuracy and Reliability**: MobileNetV3 achieved 99.66% accuracy and strong precision/recall values, proving the method's reliability for multi-cancer detection.
- **Efficient Comparison of CNN Architectures**: The study provides a clear performance comparison among multiple CNN models, showcasing MobileNetV3's efficiency for real-world use.

- **Superior Performance**: The multi-stage approach achieves state-of-the-art performance, with absolute accuracy improvements of 1.4% (ArMeme) and 2.2% (Hateful Memes).
- **Robust Preprocessing Pipeline**:Incorporated quality assessment for image sharpness and noise removal, ensuring high-quality data input for consistent model training.
- **Balanced Dataset Utilization**: Used 25,000 images across five cancer classes, enabling balanced learning and reducing overfitting.

Disadvantages

- **Limited Cancer Types**:The current implementation only covers lung and colon cancers, excluding other major cancer types.
- **Dataset Dependence**: Relies solely on the Kaggle Multi-Cancer Dataset, which may not fully represent real-world clinical diversity
- **Lack of Explainability**:The model lacks interpretability mechanisms (e.g., Grad-CAM or SHAP) for clinicians to visualize decision rationale.
- **No Clinical Validation**:The framework has not been tested in real hospital environments or through clinical trials.

2.14 Multi-Class Brain Tumor Detection Using CNN-Based Medical Imaging Analysis (2025)

Authors: Laxman Singh, Ayan Banerjee, Afshan Hassan Wani, Harishchander Anandaram, Arava Nagasri, and Bakshish Singh

Summary
This study focuses on developing a CNN-based automated system for detecting and classifying multi-class brain tumors — glioma, meningioma, pituitary, and healthy brain tissue — using MRI imaging. The researchers evaluated DenseNet169, ResNet50, VGG16, and VGG19 architectures on a combined dataset containing both online and real-time MRI images, split into 70% training and 30Among all models, DenseNet169 achieved the best results with 97.45% accuracy, outperforming ResNet50 (92.34%), VGG19 (90.23%), and VGG16 (88.65%). The DenseNet architecture's dense connectivity and feature reuse allowed it to capture subtle tumor boundaries, minimizing false positives and negatives. The paper concludes that DenseNet169 is highly suitable for real-time clinical implementation, providing rapid and accurate MRI-based tumor detection.

Advantages

- **High Multi-Class Accuracy**:DenseNet169 achieved 97.45% accuracy, outperforming other CNN models in detecting glioma, meningioma, and pituitary tumors.
- **Effective Feature Reuse**:Dense connections in DenseNet169 enable efficient feature propagation and reuse, improving learning from small datasets.

- **Robust Preprocessing Pipeline:**:Preprocessing included noise reduction, contrast enhancement, normalization, and data augmentation, ensuring reliable model training.
- **Comprehensive Evaluation Metrics**Performance was assessed using accuracy, precision, recall, F1-score, and confusion matrices for detailed model validation..

Disadvantages

- **High Computational Requirements:**:DenseNet169's dense connectivity results in high memory and GPU usage, limiting accessibility on basic systems.
- **Limited Dataset Diversity**:The dataset lacks global and demographic variety, restricting model generalization to all clinical settings
- **No Explainability Integration**:The model lacks interpretability tools (like Grad-CAM) for visualizing decision-making areas in MRI scans

2.15 Chatbots in Cancer Applications, Advantages and Disadvantages: All That Glitters Is Not Gold (2024)

Authors:Georgios Goumas, Theodoros I. Dardavesis, Konstantinos Syrigos, Nikolaos Syrigos, and Effie Simou

Summary
This paper proposes an LLM-based system for the real-time detection of social engineering attacks (e.g., phishing, pretexting) which exploit human psychology rather than technical vulnerabilities. The system overcomes the limitations of traditional rule-based and heuristic methods by continuously analyzing communication streams and building conversation history and user profiles. It uses LLMs to assess messages for style consistency, intent divergence, urgency cues, and impersonation risk. The LLM outputs feed a composite risk scoring module that enables early intervention and reduces losses, achieving higher recall and precision than traditional baselines.

Advantages

- **24/7 Accessibility and Efficiency**:Chatbots provide round-the-clock patient support, reducing response time and easing healthcare system workloads.
- **Patient Empowerment**:They encourage self-care and active participation, transforming patients into digitally literate "e-patients."
- **Scalability and Cost Reduction:**: AI-driven chatbots allow mass communication with low operational costs and minimal human intervention.
- **Personalized Interaction:**:Chatbots adapt responses based on user data and preferences, enhancing engagement and satisfaction.

Disadvantages

- **Limited Diagnostic Accuracy**:AI chatbots may generate inaccurate or misleading medical advice, posing risks in serious conditions like cancer.

- **Lack of Empathy and Human Touch**:They fail to deliver the emotional sensitivity needed for delicate communications, such as conveying diagnoses.
- **Privacy and Security Risks:**:Handling of sensitive medical data raises concerns about confidentiality and cybersecurity breaches.
- **Dependence on Internet and Technology**:Access is limited in rural or low-connectivity regions, restricting usability among vulnerable populations.

Chapter 3
PROPOSED WORK

3.1 PROBLEM STATEMENT

Existing Artificial Intelligence (AI)–based cancer detection systems exhibit several critical limitations that the proposed work — AI-Driven Multimodal Cancer Detection and Diagnostic Assistant using Vision Transformers and Fine-Tuned AI Agents — aims to overcome.

- Limited Multimodal Integration

  Most current research focuses only on a single diagnostic modality (e.g., medical imaging or blood test data). Systems that specialize in image classification do not integrate textual or clinical data, resulting in fragmented diagnostic insights. This lack of multimodal fusion reduces diagnostic confidence and fails to provide doctors with a unified, evidence-based decision support system.

- Dependence on Conventional CNN Architectures

  Traditional deep learning models such as CNNs (Convolutional Neural Networks) often struggle to capture global spatial dependencies in medical images. While effective in local feature extraction, CNNs miss long-range contextual patterns crucial for detecting subtle tumor boundaries and irregular tissue morphologies. This limits accuracy and interpretability, especially for complex cancers like lung and brain tumors.

- Absence of Domain-Specific AI Agents

  Most existing chatbots or automated systems in healthcare rely on generic Large Language Models (LLMs) that lack medical awareness. These models cannot accurately respond to cancer-related queries, explain medical reports, or provide contextual recommendations. The absence of a fine-tuned AI agent trained on oncology data prevents the creation of a meaningful, intelligent assistant for both patients and clinicians.

- Limited Use of Clinical Parameters

  Conventional AI models for cancer detection focus on image-based prediction and neglect vital numerical data from blood test reports, which contain early indicators of malignancy. The exclusion of such quantitative biomarkers reduces the diagnostic completeness of the system.

Lack of Real-Time, Integrated Deployment
While many research prototypes demonstrate high accuracy in isolated environments, few provide real-time web-based deployment. The lack of integration through a lightweight framework (e.g., Flask) restricts accessibility, usability, and scalability for medical practitioners and researchers.
This project addresses these limitations by developing a three-phase AI-driven multimodal cancer detection system, combining Vision Transformers (ViT) for medical image

classification, fine-tuned AI agents for cancer-related reasoning and report interpretation, and machine learning models for blood report analysis. The integrated web-based platform, powered by Flask and PyTorch, aims to provide an accurate, interpretable, and interactive cancer diagnostic assistant.

3.2 PROPOSED METHODOLOGY

The proposed work is structured around a three-phase multimodal methodology, implemented entirely in Python using PyTorch for deep learning and Flask for integration and web deployment. Each phase focuses on one modality of cancer diagnosis, with a unified backend architecture designed to ensure seamless performance and high interpretability.

3.2.1 Core System Goals and Performance Objectives

The success of the proposed cancer detection system will be measured based on the following key objectives:

1. **High Image Classification Accuracy:**Achieve a test accuracy of $\geq 95\%$ in Vision Transformer (ViT)–based medical image classification.
2. **Reliable Cancer Risk Prediction:** Achieve $\geq 93\%$ overall accuracy in blood report–based cancer risk classification using structured laboratory data.
3. **Conversational Intelligence through AI Agent:** The fine-tuned AI assistant should demonstrate $\geq 90\%$ response relevance and contextual accuracy for cancer-related medical queries.
4. **Integrated Multimodal System:** Combine image, text, and blood data within a unified Flask-based interface, capable of processing all modalities in real time.
5. **Model Efficiency and Explainability:**Maintain optimal model performance (GPU usage $< 80\%$) while implementing attention-based explainability for ViT predictions.

3.2.2 Multi-Stage Processing Workflow

The proposed cancer detection framework follows a sequential, three-phase processing pipeline that transforms raw multimodal data (image, text, and numeric) into structured diagnostic predictions.

1. **Stage 1: Medical Image Classification Using Vision Transformer (ViT):**
    - Medical image datasets are collected from open repositories such as Kaggle and TCIA.
    - Each medical image is divided into fixed-size patches.
    - Linear embedding transforms each patch into a vector representation for the Vision Transformer encoder.
    - Preprocessing involves resizing, normalization, and augmentation (rotation, scaling, and flipping) to improve generalization.
    - The ViT model applies multi-head self-attention to capture both local and global contextual features.

- A classification head predicts the cancer type (e.g., lung, breast, or brain) with corresponding confidence scores.
- Fine-tuning is performed using AdamW optimizer and early stopping (at peak validation accuracy).
- Accuracy, Recall, and F1-score are monitored to evaluate robustness.

2. **Stage 2: Fine-Tuning the AI Agent for Cancer-Related Tasks**

   - A pre-trained LLM (e.g., GPT-2 or BioGPT) is fine-tuned using cancer-related literature, research articles, and clinical notes.
   - Tokenization and contextual embedding are performed using Hugging Face Transformers library.
   - The model is optimized to perform tasks such as:
     - Cancer information retrieval
     - Report summarization
     - Medical question answering
   - Training uses cross-entropy loss and early stopping to avoid overfitting.
   - The fine-tuned LLM is integrated into a Flask chatbot module.
   - Users can ask cancer-related questions (e.g., "What are early symptoms of breast cancer?") and receive context-aware, medically relevant answers.

3. **Stage 3: Blood Report–Based Cancer Prediction**

   - Numerical datasets are collected from open medical repositories containing parameters such as RBC, WBC, Hemoglobin, and Platelet Count.
   - Data cleaning (handling missing values) and scaling are performed using Pandas and Scikit-learn.
   - A Deep Neural Network (DNN) is implemented in PyTorch for binary classification (Cancer/No Cancer).
   - The network is trained using ReLU activation, Adam optimizer, and Binary Cross-Entropy loss
   - The model returns a cancer probability score between 0 and 1, along with class labels.
   - Integration through Flask Web Framework
   - Flask serves as the central integration layer that connects the three models.
   - Users interact via a web-based dashboard, where they can:
     - Upload medical images for classification.
     - Enter blood parameters for cancer prediction.
     - Chat with the AI assistant for medical guidance.
   - Flask handles backend communication, REST API routing, and result visualization in real time.

3.3 SYSTEM DESIGN

The system architecture is designed as a modular, scalable, and multimodal pipeline consisting of three core layers: Input Layer, Core Model Layer, and Output and Integration Layer.

3.3.1 Input Layer Design (Conversation Processing)

Collect, validate, and preprocess input data from multiple sources (medical images, text queries, and blood reports).

Components and Requirements

- **Image Preprocessing**: Performed using OpenCV and NumPy (resize $\rightarrow$ normalize $\rightarrow$ patchify).
- **Text Preprocessing**: Tokenization, sentence segmentation, and stop-word removal.
- **Numerical Data Cleaning:** Handling missing or abnormal entries using Pandas.

Output

Preprocessed tensors and structured data ready for input into the Core Model Layer.

3.3.2 Core Model Design (Deep Learning and NLP Architectures)

Components and Requirements

- **Purpose:** Perform high-accuracy cancer detection through AI models tailored for each data type.
- **Base Architecture:** Vision Transformer (ViT) for image-based cancer classification,Fine-tuned Large Language Model (LLM) for cancer text reasoning,Neural Network Classifier for blood test analysis.
- **Optimization Strategies:** Pre-trained ViT and LLM weights are fine-tuned for domain adaptation, Weighted cross-entropy loss is used to manage class imbalance,Training halts automatically when validation loss stops improving (typically at Epoch 3–5).

3.3.3 Output and Analysis Layer (Result Generation and Visualization)

Components and Requirements

- **Purpose:** Interpret model outputs, visualize predictions, and provide real-time diagnostic insights through the Flask web interface.
- **Components:**Flask Dashboard: Displays predictions with labels and confidence scores,Visualization Tools: Attention heatmaps (for ViT) highlight tumor areas,Chatbot Interface: AI assistant provides explanations or next-step recommendations,Performance Monitoring: Real-time logs display accuracy, F1-score, and latency metrics.

- **Evaluation Metrics:**Confusion Matrix for multi-class image classification, Precision–Recall and ROC curves for blood-based predictions,Response Coherence Scores for LLM chatbot performance.

Chapter 4
IMPLEMENTATION AND RESULTS

4.1 System Workflow Overview

This chapter details the implementation of the AI-assisted multi-cancer detection system, highlighting its workflow, technical components, and user interaction design. The system integrates medical imaging, report parsing, and AI-driven guidance to provide a holistic diagnostic support tool.

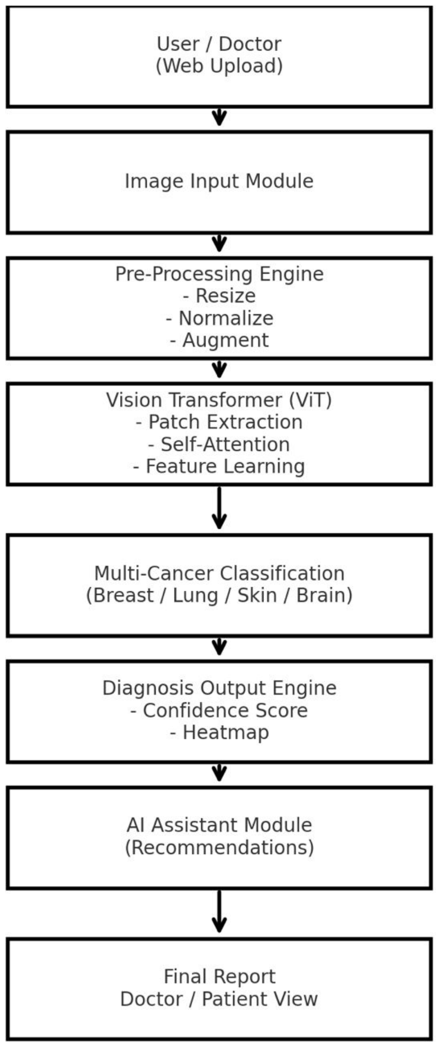

Figure 4.1: System Workflow Flowchart

4.2 User Uploads Medical Image & Medical Reports

Description

In the initial stage, users—either doctors or patients—upload cancer-related medical images and supporting documents. The system accepts:

- **Medical Images:** Histopathology slides, CT scans, MRI scans, dermoscopy images (JPG, PNG, DICOM).
- **Medical Reports:** Blood reports, biopsy results, pathology findings, treatment summaries (PDF, JPG, PNG).

The inclusion of both images and reports enables the AI to provide context-aware, medically relevant insights. The interface is designed for simplicity and accessibility, ensuring ease of use for all stakeholders.

4.3 Pre-Processing of Images & Medical Report Parsing

Image Pre-Processing

Uploaded images undergo:

- Resizing to a standardized resolution.
- Normalization for consistent pixel value distribution.
- Noise removal to enhance feature clarity.

Medical Report Parsing

Medical reports are processed using:

- **OCR:** Text extraction from scanned or image-based reports.
- **NLP:** Identification and extraction of key medical parameters (e.g., WBC count, tumor grade, lesion measurements, biopsy notes).

This dual processing ensures both visual and textual clinical findings contribute to the system's diagnostic understanding.

4.4 Vision Transformer Extracts Cancer-Related Features

Feature Extraction

The Vision Transformer (ViT) processes the medical image by:

- Splitting the image into patches.
- Applying self-attention layers to analyze visual cancer markers.
- Identifying structural abnormalities, tissue texture changes, cellular deformation, and tumor signatures.

ViT's ability to combine global and fine-grained features enables accurate detection of early-stage cancers across multiple modalities, making the system versatile for multi-cancer detection.

4.5 Model Predicts Cancer Type & Uses Report Data

Prediction Process

The model classifies the input as a specific cancer type or non-cancerous case, providing:

- Softmax probabilities and confidence scores.
- Cross-referencing with extracted medical report data (lab values, textual notes).

This hybrid approach improves reliability and reduces false predictions, bridging the gap between image-only systems and real clinical workflows.

4.6 AI Assistant: Q&A & Medical Guidance

Interactive AI Support

Users can interact with the AI assistant to ask questions about:

- Cancer symptoms and lab results.
- Medical terms and treatment guidelines.
- Interpretation of test results.

Example Queries:

- "What does 'adenocarcinoma grade III' mean?"
- "What further tests should be done after a CT scan showing a mass?"
- "Explain my biopsy result."

The AI provides accurate, patient-friendly explanations and general guidance, acting as a medical companion. **Note:** The system offers general AI assistance; final medical decisions remain with qualified doctors.

4.7 Final Diagnosis Report & Review

Diagnostic Interface

The system presents a comprehensive diagnostic interface, including:

- Probable cancer type and confidence level.
- Highlighted attention/heat-map regions in the image.
- Summary of medical report findings.
- AI-powered explanations and suggestions.
- Downloadable report for doctors and patients.

Doctors verify AI-assisted insights and provide final conclusions, while patients gain clearer understanding, reducing anxiety and confusion through guided explanations.

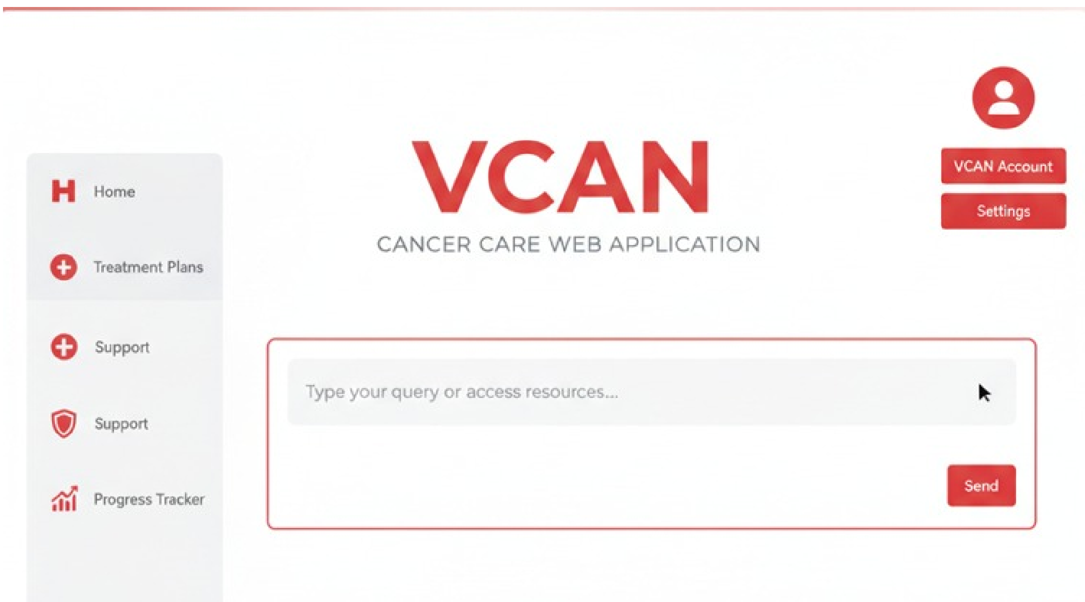

Figure 4.2: Sample AI-Generated Diagnosis Report

Chapter 5
CONCLUSION

The project titled "AI-Driven Multimodal Cancer Detection and Diagnostic Assistant" aims to create an intelligent system that assists in the early detection and analysis of cancer using Artificial Intelligence (AI) and Deep Learning. The system is designed around three core components — Vision Transformer (ViT) for medical image classification, fine-tuned AI agent for cancer-related reasoning, and blood report analysis for cancer prediction — all integrated into a unified, web-based interface using Flask and Python.

At this stage, the User Interface (UI) has been fully designed and developed using Flask, HTML, CSS, and JavaScript. The web platform provides users with the ability to:

- Upload medical images for classification,
- Input blood report parameters for prediction, and
- Interact with an AI chatbot for medical queries.

In addition to the UI, the data collection phase for both medical images and blood sample datasets has been successfully completed. The image datasets consist of cancer-related samples (such as lung, breast, and brain cancer images), which will be used to train the Vision Transformer (ViT) model. Similarly, blood test datasets have been collected, containing parameters such as RBC count, WBC count, hemoglobin levels, and platelet counts, which will be used for training the deep learning–based blood analysis model.

These achievements mark a significant milestone in the project's progress — establishing both the frontend framework and the data foundation for AI model training and integration. Once trained and optimized, these models will be incorporated into the existing Flask framework to enable full automation of cancer detection, report analysis, and medical query handling.

5.1 FUTURE WORK

The upcoming development phase will focus on AI model training, backend integration, and performance evaluation. The detailed roadmap for the next phase includes the following tasks:

5.1.1 Preprocessing and Augmentation of Collected Datasets:

- Clean and preprocess the collected medical image and blood sample datasets.
- Apply normalization, resizing, and augmentation techniques to improve model performance and generalization.

5.1.2 Vision Transformer (ViT) Model Training:

- Train the ViT architecture using the collected image dataset for multi-class cancer classification.

- Fine-tune the model using PyTorch to achieve high classification accuracy and interpretability through attention maps.

5.1.3 Blood Report Classifier Development:

- Train a Deep Neural Network (DNN) or Random Forest model using the collected blood data.
- Evaluate the model's performance using metrics such as accuracy, precision, recall, and F1-score.

5.2 Fine-Tuning the AI Agent (LLM):

- Fine-tune a pre-trained model such as BioGPT or GPT-2 on cancer-related text corpora.
- Integrate the model into the chatbot interface for intelligent question answering.

5.3 Integration with Flask Application:

- Connect all trained models (ViT, LLM, and Blood Classifier) to their respective Flask routes (/predict_image, /predict_blood, /ask_ai).
- Enable real-time inference and result visualization on the web UI.

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